Haasith Venkata Sai Pasala

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Education

International Institute of Information Technology Hyderabad (IIIT-H) Hyderabad, India
M.S. by Research in Electronics and Communication Engineering (Robotics) *Aug. 2021 – Jun. 2024*

Anil Neerukonda Institute of Technology and Sciences Vishakapatnam, India
Bachelor of Engineering in Electronics and Communication Engineering *Aug. 2014 – May 2018*

Technical Skills

Languages: C/C++, Python, Java, JavaScript, C
Autonomy & Perception: ROS 1 / ROS 2 (Foxy, Humble), Nav2, SLAM (FAST-LIO, D-LIO, RTAB-Map), OpenCV, PointNet++, Sensor Fusion (GPS, Lidar), Visual Servoing, Trajectory Optimization
Simulation & Tools: Docker, AWS EC2, Gazebo, PyBullet, Git, CMake, Linux
Embedded & Hardware: Embedded Linux, ARM Cortex, ALSA, I2C, SPI, Unitree Go2W, XArm7
Testing & QA: Appium, Selenium, Cucumber (BDD)

Experience

Technical Lead (Consultant) Jul. 2025 – Dec. 2025
Stealth Mode Company *Remote*

- Engineered the complete autonomy stack and environment setup for a Unitree Go2W quadruped robot, migrating the control architecture between ROS 2 Foxy and Humble to ensure hardware driver compatibility.
- Designed and integrated a robust state estimation pipeline utilizing FAST-LIO, D-LIO, and RTAB-Map for precise localization, fusing GPS data for global positioning in outdoor environments.
- Configured Nav2 and implemented whole-body control frameworks to enable autonomous navigation, obstacle avoidance, and complex legged locomotion.
- Containerized the full simulation and deployment stack using Docker and provisioned AWS EC2 instances for scalable, cloud-based testing prior to hardware deployment.

Embedded Software Engineer Jun. 2024 – Jun. 2025
Miko *Mumbai, India*

- Re-architected the ”Miko Dance” feature for offline execution by replacing WebSocket-dependent network calls with localized C++ processing logic, optimizing system responsiveness.
- Led hardware-software system bring-up and validation, systematically debugging critical latency, synchronization, and audio quality anomalies across the entire tech stack.
- Developed and integrated ALSA-based audio subsystems on ARM embedded Linux, enabling highly reliable audio interaction for production-level consumer robots.
- Implemented low-level C/C++ software drivers to configure the TAS2563 audio amplifier, managing communication via I2C and I2S/TDM interfaces.

Graduate Research Assistant Sep. 2021 – Jun. 2024
Robotics Research Centre, IIIT-Hyderabad *Hyderabad, India*

- Engineered an advanced perception pipeline for throwing manipulation by deploying PointNet++ for 3D point cloud analysis and integrating ensemble learning (SVM/Random Forests) for robust object recognition.
- Formulated rigid body dynamics and implemented trajectory optimization algorithms using MATLAB, Gazebo, and ROS to compute optimal initial conditions for dynamic object throwing.
- Designed and integrated a novel hybrid end-effector with an XArm7 robotic arm and mobile base, developing the control architecture using ROS, Raspberry Pi, and Arduino.
- Conducted two-stage parameter identification and applied optimization techniques to bridge mathematical models with physical hardware, drastically minimizing sim-to-real discrepancies.

Consultant Engineer Mar. 2019 – Jun. 2020
L&T Technology Services *Mysore, India*

- Engineered end-to-end automation testing frameworks for a regulated mobile medical application (Eli Lilly) across two major release cycles, utilizing BDD principles with Cucumber.
- Designed and executed cross-platform automated testing pipelines for iOS and Android environments (real devices and emulators) utilizing Appium, WebdriverIO, and Selenium.
- Programmed scalable test features and step definitions in Java (TestNG) and JavaScript, ensuring high reliability and strict compliance for a critical healthcare application.

Projects

Stereo SLAM System Development | *C++, OpenCV, Pose-Graph Optimization* Sep. 2021 – Dec. 2021

- Implemented Stereo Reconstruction, Perspective-n-Point (PnP), and Bundle Adjustment to develop a robust 1D and 2D SLAM pipeline.
- Applied Pose-Graph Optimization algorithms to correct trajectory drift and significantly enhance overall localization accuracy.

Design of an Anthropomorphic Underactuated Hand | *MuJoCo, PID Control, C++* March 2022

- Engineered a multi-fingered anthropomorphic robotic hand in the MuJoCo physics engine, implementing underactuated mechanisms to achieve human-like grasping synergies.
- Designed and tuned a decoupled PID control architecture for individual finger actuation, optimizing for both precision fingertip manipulation and power grasps.
- Developed a kinematic mapping pipeline to translate human hand poses into robotic joint commands, validating the system's ability to perform complex, non-prehensile tasks.

Eye-in-Hand Visual Servoing for Dynamic Tracking | *OpenCV, ROS, C++/Python* Jun. 2023

- Developed an active visual servoing control loop using an XArm manipulator equipped with an end-effector-mounted RGB-D camera.
- Utilized OpenCV and ROS TF to estimate the 6D pose of ArUco markers, computing the necessary velocity commands to dynamically adjust the robot's trajectory in real-time.

GELLO Framework for XArm Teleoperation | *Robotics, Kinematics, ROS* Nov. 2023

- Led the integration and deployment of the GELLO teleoperation framework for an XArm manipulator, architecting the end-to-end system for robust data collection and imitation learning.
- Configured a low-latency leader-follower control system, ensuring precise joint-space mapping and real-time responsiveness for complex remote manipulation tasks.

Publications & Recognition

Identification and Learning-Based Control of an End-Effector for Targeted Throwing | *IEEE RA-L*Aug. 2024

- Accepted to the IEEE Robotics and Automation Letters (RA-L).
- Featured on **IEEE Spectrum** (Oct. 2024), highlighting the multimedia extension and real-world throwing manipulation capabilities.

A Novel Hybrid Gripper Capable of Grasping and Throwing Manipulation | *IEEE/ASME T-MECH* 2023

- Published in the IEEE/ASME Transactions on Mechatronics (Impact Factor: 6.4).
- Featured on **IEEE Spectrum** (Apr. 2023) for pioneering a dual-action mechanism bridging grasping and dynamic throwing.